

Paving the way towards a methodology to faithfully assess physical, physiological, and cognitive impacts of Augmented Reality under constrained environments: a Head-Mounted Display use case

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ABSTRACT

Devices supporting augmented reality are becoming more and more advanced and increasing attention is being paid to this technology. Due to recent upgrades in technology, the quality of the experience has improved for users, especially the spatial and temporal consistencies of the scene displayed, and scientists' efforts are focused on designing innovative and seamless interaction modalities, especially in the aviation sector. Nevertheless, the level of requirements being remarkably high in such industry, the implementation of this technology raises the question of the impact of its use under constrained environments and how scientific validation of devices can be performed. Therefore, it is needed to implement a methodology to assess the effects of augmented reality on an operator in a constrained environment. Firstly, the challenges that arise when assessing augmented reality impacts under constrained environments are discussed by emphasizing on the specific use case of head-mounted displays. Then, the proposed methodology that can be used to overcome these challenges is introduced. The methodology involves assessing the impact of augmented reality both objectively and subjectively using metrics that are specific to the application. Finally, recommendations for future research as well as ways of improvement of the current methodology are also provided. The results and the validity of the methodology will be the object of a future article.

CCS CONCEPTS

• Human-centered computing; • HCI design and evaluation methods;

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KEYWORDS

Methodology, augmented reality, Head Mounted Display, Human factors, Eye-Tracking, physical effects, physiological effects, cognitive effects

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1 INTRODUCTION

The emergence of Augmented Reality (AR) technologies offers new perspectives in different domains of the aeronautical industry (design process; maintenance; training; implementation of functionalities in flight) [1–4]. Nevertheless, the aeronautical context is associated with reliability requirements for which it is imperative to master all aspects of a technology before its implementation. The use of these devices raises several questions, notably to ensure that the principles and symbology implemented will not be detrimental to operational performance. Ling et al. (2013) found that a combination of adverse effects arises in 80-95% of individuals using stereoscopic headsets [5]. Assessing the impact of augmented reality is a challenging task, especially under constrained environments.

The aim of this paper is to propose a methodology to assess the effects of augmented reality in a constrained environment with a specific focus on head mounted displays. Section 2 introduces the known physical; physiological and cognitive effects of AR. Section 3 present the usual methodology to assess these effects for Head Mounted Display (HMD) and Section 4 details four experimental protocols to evaluate objectively and subjectively the physical, physiological, and cognitive effects of an HMD, in a constrained environment. The purpose is not to provide an exhaustive review of each of these complex effects, but to consider the results of previous work in these areas in an attempt to understand how to assess the

possible effects of HMD and propose a methodology that could be applied under different environments to quantify such impacts. The results and the validity of the methodology will not be discussed in this article.

2 KNOWN EFFECTS OF AR

When it comes to displaying AR information, a wearable device such as an HMD is a commonly used means; however various issues such as head-supported weight, center-of-mass offset, imbalance, inertial effects, pressure points, sweating, and heat development can negatively impact the user's activity, comfort, and acceptance of the device. Extreme head positions can also cause joint reaction forces, stress passive tissues and structures, and occlude blood flow [6]. The compression stress caused by carrying a load on the top of the skull can also result in various symptoms in the neck, lower back, upper back, and shoulders. High vibration or buffeting will exacerbate the effect of this extra weight and displaced center of mass, with effects that can range from fatigue and neck strain to serious injury [7].

A common physiological adverse effect of mixed reality is cybersickness, it is a cluster of symptoms experienced by users during or after exposure to an immersive environment [8], characterized as a physiological response to an unusual sensory stimulus similar to motion sickness [9]. It entails different symptoms varying from a mere feeling of unpleasantness to strong nausea and vomiting [10]. Although virtual reality and augmented reality technologies have improved in recent years, cybersickness remains a common issue reported by users [11]. The incidence and degree of intensity of the symptoms vary, but the duration of the symptoms remains stable [12, 13]. The severity of cybersickness increases with exposure time. Eye strain is a common symptom of cybersickness, particularly prevalent in AR use, and can significantly impact users' activities. Eye strain, or asthenopia (literally "eye without strength" Mackenzie, 1843) is an eye condition that manifests through non-specific symptoms such as fatigue, pain in or around the eyes, blurred vision, headaches, dry eyes, and occasional double vision.

According to Melzer [14], one of the purposes of designing AR systems into an HMD is to go beyond a hands-free display to a point where it can be considered a cognitive prosthesis, assisting the user in enhancing situation awareness, encouraging, or enabling correct decision-making and reducing workload. Nonetheless, it is important to be cautious and avoid any potential cognitive adverse effects that may be triggered by display properties, such as visual clutter. This concept is defined by Doyon-Poulin [15] as: "The state of a display exhibiting at least one of the following three conditions: (1) an abundance or variety of symbols or colors, (2) information deemed irrelevant to the task and context, (3) a lack of organization and structure." AR can exacerbate clutter by contextually linking information to the objects it describes, which limits where it can be placed on the display [16], with items close to each other interfering with their mutual visibility [17]. Visual clutter can cause occlusion or fragmentation of information and its effects are exacerbated by stress and workload [18]. Clutter can hinder event detection and reduce visual acuity, especially in the peripheral region [19, 20].

A cognitive adverse effect that is worsened by visual clutter is attention switching. Huckauf et al. [21] discovered that switching between virtual and real-world information led to decreased visual performance. McCann, Foyle & Johnston [22] showed that pilots

have difficulties processing HUD-referenced information and world-fixed information simultaneously, transitioning from HUD to the environment can cost up to 150 ms due to an attention shift. Shifting from virtual to real-world information aggravates this effect as AR symbology captures more attention and can be associated with attentional tunneling which refers to the locking-in of a hypothesis or goal for an extended period of time, resulting in the neglect of other information and tasks. It can cause operators to focus excessively on irrelevant information and disregard critical information such as alarms [23–25]

A users' behavior is the result of decision-making, which depends on their situation awareness (SA) – understanding the current and future state of their environment. AR can enhance SA by directing attention and inferring information. Scientists found that AR can improve drivers' SA of surrounding vehicles and pedestrians, facilitating decision-making. However, over-reliance on AR can be problematic if the information is inaccurate or not understandable [26, 27].

Since the 1970s, Mental Workload (MW) has been extensively discussed and researched [28–30]. According to Gopher and Donchin [31], workload refers to aspects of the interaction between a person and a task that exceed the individual's ability to perform. By integrating information on the environment, augmented Reality has been shown to reduce workload in various fields, including aviation, manufacturing, and healthcare [32–34], with HMDs reducing MW even more than HUDs, as the information displayed does not require additional effort from the user to coordinate their field of view and the information provided [35].

To anticipate the problems that could be caused by AR, we will explore its potential harmful effects by focusing on the particular case of HMD. See-through HMDs are becoming more advanced, and attention is focused on designing innovative interaction modalities [36], especially in the aviation industry which is the 4th industrial sector in terms of number of scientific papers on AR [37] HMDs offer real-time information directed seamlessly within the user's field of view. It has been shown to reduce perceptual switching times, workload, and improve situation awareness for users performing various tasks [38–40]. As a wearable device, the HMD poses challenges in evaluating its impact on physiological and cognitive factors and necessitates specialized scientific approaches due to its unique characteristics.

3 HOW TO ASSESS THESE EFFECTS FOR AN HMD

3.1 Physical stress and comfort assessment

The measurement of discomfort itself is a fundamental difficulty. Discomfort is a generic and subjective sensation that arises when human physiological homeostasis, psychological well-being, or both, are negatively affected. In order to gather operator's feelings and biomechanical / musculoskeletal reactions to see-through HMD exposition, objective and subjective measures are combined. Additional head-borne mass has been linked to higher muscle activity, as measured with electromyogram [EMG; 41]. Surface EMG recording to the cervical region of the spine could hence provide a prediction of muscle fatigue induced by an HMD. Motion sensors can also be used to compare the amounts of head movements and posture change with and without HMD. Questionnaires can provide critical insight into perceived device-induced discomfort. Multiple

studies [42, 43] used Borg's scale to evaluate subjective discomfort when wearing an HMD. The Comfort Rating Scale was developed for wearable computers [44]. The tool is suitable for evaluating the comfort of body-attached items [45]. Developing an interview guide related to comfort can validate data from standardized scales while providing more information on participant comfort [46].

3.2 Physiological effects assessment

Objective measures of cybersickness such as postural sway, electrocardiogram, electrogastrogram, galvanic skin response or electroencephalogram are interesting but requires invasive sensors, and may induce various biases and technical issues, especially in the context of operational *out of the lab* experiment [47, 48]. Moreover, their implementation may be unnecessary given that the Simulator Sickness Questionnaire can provide detailed results and can be decomposed into clusters [SSQ; 49]. Subjective questionnaires can also complement objective measures of visual fatigue [49]. Oculomotor class symptoms included in the SSQ questionnaire can serve as a reliable metric for visual discomfort.

Eye tracking data can supplement the SSQ but cannot be an indicator of cybersickness symptoms on its own as it only correlates with it [50–52]. However, the eye blink is a well-known indicator of visual fatigue [53, 54]. Indeed, a blink decrease contributes to poor tear film quality and temporarily stresses the cornea causing dry eye, which is a common symptom of Computer Vision Syndrome (CVS) [55, 56]. Hence, Blink rate per minute can be used as a predictor of asthenopia using standard eye tracking devices [57]. Visual fatigue can also be measured using accommodative parameters especially critical flicker-fusion frequency (CFFF) [58], which measures the point at which a flickering stimulus appears to be continuous to an observer's eyes. Decreases in CFFF values indicate decreases in alertness and visual perception function [59, 60], and a simple setup can be used to measure it, making it a popular option [61].

3.3 Measuring the visual clutter

There is an agreement that clutter may represent a significant problem to users of AR HMDs. However, definitions of clutter vary widely [15], which explains the range of measurement approaches, including image-processing algorithms, performance evaluation, subjective evaluation, and eye tracking. For image processing algorithms, a program is used to determine the amount of clutter on a display based on properties such as pixel density, lay-out, and target-background similarity: One common clutter measurement technique is to combine feature congestion, edge density, and sub-band entropy to provide an overall estimate of clutter [63]. An objective clutter indicator for AR display can be established by measuring the ratio of active pixel on an HMD [64]. Subject matter expert can be asked to rate display clutter with questionnaires using the Clutter Rating Scale, they are first asked to rank dimensions of visual clutter by order of contribution to visual clutter in the context of the activity performed during the experiment and then give a rating of the display on each dimension. [65, 66]. Moacdieh and Sarter [66] studied multiple eye-tracking data to measure clutter and found twelve that correlated with the level of clutter displayed. However, the use of these metrics requires specific protocol with target definition that is more suited for laboratory context than constrained operational environments.

3.4 Human attention assessment

The coupling between eye movements and attention has been well-documented, showing that attention is deployed to the location of a future saccade even though restrictions have been elicited [67–69] making eye tracking measurements a suitable candidate to predict attentional state [70, 71]. Fixation length and frequency indicate information processing complexity [72]. Changes in fixation positions during visual search reflect changes in attention, and saccade range reflects the width of attention [73]. Counting gaze shifts between Areas of Interest (AoI) and fixations in each of them can contribute to assessing attention shifting [74].

"Visual tunnelling" is a manifestation of attention tunnelling that can be observed through gaze behavior. Attentional focus and its extreme manifestation, attention tunneling, are marked by decreased saccadic activity and prolonged fixation on specific points, resulting in fewer areas of interest being scanned [75, 76]. This spatial gaze concentration as well as more frequent and longer fixations away from the primary task can affect the detection of potential hazards [77, 78].

The detection –or not– of an unexpected event or object near the primary task may constitute an additional assessment of the attention's distribution and the level of focus on one task instead of two, in this context, gaze analysis can be a complementary tool for the understanding of the detection rate of visual alarms [24, 25].

3.5 Situation awareness & mental workload

While the effects of AR on SA and MW have been extensively researched, measuring these effects requires a highly specific protocol strongly linked with the task investigated. Standardized scales can be used as a subjective comparison between scenarios performed with and without the AR device investigated. Situational Awareness Rating Technique, [SART; 79] is a standardized scale used to measure SA. NASA TLX [Task Load Index; 80] is one of the most used MW standardized scales [81], which makes it convenient to compare with other studies performed in the aeronautical field. While physiological measures such as electrodermal response or ECG have been used to assess SA and MW, there is no consensus on their effectiveness and the issue of confounding variables with stress and physical activity remains. eye tracking measures may offer a promising means of evaluating SA and MW, but in our study, they will only be used for exploratory purposes. Performance metrics shall be designed with respect to the expected operator's behavior on the ongoing task. Since the aim of the present methodology is to investigate impacts of AR under various constrained environments, this will not be deeply detailed within this document.

3.6 How to assess those effects in a constrained environment

Conducting an experiment to assess the impacts of a see-through HMD hereabove mentioned in a constrained environment presents strong challenges. One significant challenge is the integration of sensors because the electromagnetic fields that may be present in the environment can affect the accuracy of the sensors, leading to erroneous data. Space constraints are also a concern, as the presence of the HMD and sensors can limit the movement of the participant, potentially impacting its ability to perform tasks. Furthermore, the presence of other equipment or individuals in the environment may interfere with the data collection process or even pose security

risks. Finally, each selected sensor should adapt to the HMD device investigated. These challenges must be carefully considered and addressed to ensure the accuracy and security of the experiment.

4 MATERIALS AND METHODS

The effects that have been considered for the current use-case were chosen based on three criteria: the probability of occurrence of the effect, the severity of the effect on the activity and the assessment feasibility regarding the constrained environment of the evaluation campaign. To evaluate these effects, we also had to select appropriate sensors that would provide accurate and reliable data.

Based on this work, four distinct protocols have been elaborated for our evaluation campaign which has been fully developed on 6 operators: the physical stress and comfort protocol, visual fatigue protocol, visual clutter protocol, and visual exploration strategy and attention protocol. Each of these protocols are detailed in the following paragraph. In addition, other measurements such as mental workload, situation awareness, and system usability were also evaluated.

In order to objectively evaluate these effects, three operational scenarios were then proposed: one tutorial scenario, which allows the subjects to familiarize with the HMD, and two operational scenarios of equivalent difficulty that have been designed by subject-matter experts which will be performed respectively with and without the HMD.

The following table provides the synoptic of the experiment campaign and explains how the measures and the four protocols were combined and used with the description of their dependent variables. It should be noted that the independent variables (condition with or without the HMD) have been counterbalanced in the protocols and scenarios to avoid any biases. Before each test, a questionnaire is proposed to identify the profile of each participant as well as a final debriefing interview in order to obtain subjective data.

4.1 Physical comfort

The study requires non-invasive, small, high-precision sensors. The TEA sEMG sensors meet these requirements with dimensions of 52mm x 25mm x 14mm, a mass of 20g, and a sampling frequency of 2048 Hz. The signal is exported for analysis by the CAPTIV L-7000 software. CAPTIV analysis software is compatible with T-sens motion sensors that are also used in the study, which facilitates data synchronization.

The sternocleidomastoid (SCM) and superior trapezius (TRAP) muscles are targeted because they are located shallowly enough in the neck region to allow for reliable detection via sEMG and are less prone to interference from crosstalk. sEMG relies on a reference, typically a maximum voluntary contraction (MVC), to compare muscle activity. However, MVCs can be painful and dangerous, particularly for complex muscle groups around sensitive articulations [82], hence it is not performed. However, the experimental design limits the lack of EMG standardization: sensors remain fixed between the two conditions and comparisons between conditions are from matched groups, allowing for consistent comparisons of bio-electric activity between samples. Since the relationship between HMD parameters and physical workload can be affected by posture [83], six positions were determined to cover relevant and important neck inclinations when using the HMD: In the sEMG test, isometric contraction are used for easier interpretation of muscle fatigue [84]. Participants hold each position for 30 seconds and return to a neutral posture between positions. They keep their arms relaxed vertically along the body and avoid using their upper limbs throughout the experiment according to SENIAM recommendations [85]. Participants have to perform this test with and without HMD, conditions are counterbalanced to minimize any complications related to the order of exposure. The dependent variable retrieved are the RMS – Root Mean Square of the EMG data [86] as well as the responses to standardized scales for each posture and for each condition.

Table 1: synoptic diagram of the experiment campaign

Step	Experiment phase	Protocol concerned	Dependent Variable
1	Pre-brief	Pre-test questionnaires	SSQ score, questionnaire answers
2	Pre-scenario experiments	Physical comfort protocol Visual fatigue protocol	sEMG RMS, VAS score, answers, CFFF reference, VAS score, oculomotor SSQ items, answers
3	Tutorial scenario	Visual clutter protocol	Clutter subdimension ranking
	First operational scenario	Visual fatigue protocol	CFFF, VAS score, oculomotor SSQ
4	Second operational scenario		
5		Visual clutter protocol (IF HMD) Visual strategy/attention protocol	Clutter rating scale score Eye-tracking + motion data + rate detection of unexpected events
		Situation Awareness	SART score
		Mental Workload	NASA-TLX score
		System Usability (IF HMD)	SUS Score
6	Post scenario experiments & debrief	Physical comfort protocol, semi-direct interview	sEMG RMS, VAS score, answers,

4.2 Visual fatigue

A circuit was designed for the CFFF test on an Arduino Uno board (Arduino.cc, Italia), consisting of a LED, a potentiometer to adjust light intensity, a push button for input, and USB communication for result transmission.

A boundary method consisting of an ascending test (20Hz - 100Hz in 1Hz steps) followed immediately by a descending test (100Hz - 20Hz in 1Hz steps) is used to find the CFFF value [61], the average of both tests is retained. The participant must indicate the precise moment when he perceives the LED changing from flickering to continuous state and vice versa. The participant have to perform the CFFF test and answer the standardized scales (SSQ + Visual Analogue Scale) on three occasions: Prior to the "operational" scenario, this measure is considered as the reference measure and after each "operational" scenario: the dependent variable retrieved are the CFFF before and after each condition (in Hz), the responses to the "oculomotor" item of the standardized scale for each condition and VAS as well as the blink rate obtained via oculometric data.

4.3 Visual Clutter

The clutter measurement algorithm of Rosenholtz *et al.* [62] was originally a Matlab (Mathworks, USA) script that takes an image and returns feature congestion and subband entropy scalar. It was modified to accept videos as input, compute both scalar as well as edge density every five frame and store the values in a csv file with an image index. To measure active pixel ratio, a specific Matlab script was developed to compute the number of opaque pixels per frame.

After tutorial scenario including the display of HMD symbology, participants rank ten dimensions by their contribution to clutter for the performed task. When performing operational scenario with HMD, the symbology is recorded for analysis and participants rate clutter using standardized scales for overall impression and each dimension on two display configurations. A debrief interview is performed after both scenarios to ask specific questions about visual clutter. The video input for the clutter measurement is the Point of View (PoV) of the user recorded by the eye-tracker scene camera and the AR content from the HMD. A video editing process allows to superimpose the AR content on the PoV video. The dependent variables retrieved are Rosenholtz *et al.* global clutter score computed by the algorithm that allows to evaluate the impact of the AR content on the visual clutter of a natural scene, the Clutter Rating Scale score which is divided in two parts: the clutter subdimension ranking and the rating on each dimensions and finally the Active Pixel Ratio which is the percentage of active pixel over transparent one on every frame of the recorded symbology.

4.4 Visual exploration strategy & attention

To measure eye parameters with and without an AR-HMD, certain limitations must be considered. First, chosen instruments must provide appropriate measurements with sufficient spatial and temporal resolution. Second, to ensure metrological consistency similar tools must be used in both conditions for a rigorous comparison. Using distance-based eye-trackers is not recommended due to the HoloLens visor's tint, opacity, and reflective properties, which can

cause distortion and occlusion of the wavelength used for tracking. The eye tracker embedded in the HoloLens has a low temporal resolution (30Hz) and does not measure blink rate or pupillary dilation. The Pupil Labs 120 Hz Binocular glasses perform with an accuracy of 0.84° and a precision of 0.16° and are a valid choice when mid-range accuracy is sufficient, lab conditions are met, and repeated calibration is possible [87]. Opting for Pupil Labs solutions ensures metrological consistency between the two conditions. Indeed, the same system can be used embedded on a glasses frame for the non-HMD condition and can be adapted to fit the HoloLens 2 [88].



Figure 1: (LEFT) Pupil Core by Pupil Labs. (RIGHT) Pupil Labs adapted for the HoloLens 2 From Drouot *et al.* [88]

Head movements must also be tracked, including the number of switches between displays and eye-head coordination. TEA T-Sens Motion, which combines accelerometers, gyroscopes, and magnetometers, can measure head rotation angle values around three anatomical axes. It has a sampling rate of 64 Hz, dimensions of 60 mm x 35 mm x 15 mm and weighs 32g. It can be synchronized with up to fifteen other IMUs for full body tracking and can be used in both conditions with and without the HMD, fixed to the vertex with a harness or strap.

The dependent variables retrieved during both operational scenario for assessing visual exploration strategy and attention are the following : Measure of attention shifting especially the number of head shifts between displays, measured by cross referencing motion sensor and eye-tracking data, the number and time of fixation on each display, measurement of inattentive blindness using the rate of detection of an object verbally stated by the operator combined with eye tracking data as a complementary tool to understand the detection [89] and the analysis of visual strategy using eye-tracking data such as visual path length, average fixation time, average distance of saccades and the coordination of the eyes and the head using entropy analysis

5 RESULTS AND DISCUSSION

In the next section, we will discuss the limitations of the methods and sensors used for each effect of AR. This will include feedback, cautionary notes, and guidelines from both the literature and dry runs conducted in a constrained environment.

Concerning physical stress and comfort assessment, placing EMG sensors on subjects can be challenging as the electrodes may be unstable due to clothing, facial hair, or sweating caused by the warm temperature of the environment. In order to mitigate this issue, additional medical straps can be used, but these can cause

discomfort and are difficult to remove post-experiment. Also, removing and replacing the sensors on the electrodes during the test is not recommended as it is painful for the participants and may have an impact on data reliability. Due to the challenges associated with sEMG measurement, it is necessary to visualize data in real-time to detect any possible artifacts or sensor failure. Finally, one must be cautious when interpreting rating-scale. Indeed, despite being a convenient, and widely used approach for discomfort assessment, it is known that participants can have difficulty making a direct judgement of sensations, especially without any standard contrast stimulus. Furthermore, ratings can easily be biased due to instructions, or context [90].

When assessing visual fatigue, it must be noted that the duration of the experiment should be controlled for, as time has a significant effect on this metric [91, 92]. It is important to mention that environmental factors should also be monitored and controlled, as low ambient humidity, high forced-air heating or air conditioning settings, excess static electricity, or airborne contaminants influence eye dryness [93, 94]. For the CFFF test, participants reported discomfort during the experiment, citing the insufficient distance between their eyes and the LED, which required them to squint. They also experienced difficulty focusing on the signal due to ambient light. To address these issues, an eye box could be designed above the LED to enhance participant comfort and focus. Finally, the measure of blink rate can be influenced by other effects than visual fatigue including workload (or at least the complexity of the visual task), display type and attention [95].

Regarding the use of eye-tracking technology, differentiating between virtual and real-world fixations is not feasible with the current equipment. It is also challenging to detect peripheral vision usage, and the use of visual markers to map gaze over Areas of Interest (AoIs) was not possible due to space limitations and differences in luminosity. To overcome this issue, AoIs were defined by cross-referencing motion sensor data with eye-tracker data. Eye-tracking technology also presents reliability challenges, exacerbated by the presence of glasses or eye makeup, which can affect calibration quality. Additionally, it is recommended to avoid using automatic and adaptive camera parameters, including shutter speed and ISO, to reduce interference and lag caused by rapid infrared flashes emitted by the HoloLens embedded eye-tracker.

For MW, the main drawback of using self-reported measures is the inability to provide real-time and continuous information, which is critical given the dynamic nature of MW changes throughout a task [96]. These measures rely on participants' self-report, collected after the task is completed or after pausing the task for the report to be made, and are susceptible to a variety of factors, including individual differences in personality traits and cognitive skills [97, 98]. For subjective rating of SA on the other hand, research has shown that they correlate poorly with objective SA measures, indicating they more closely capture a person's confidence in his or her SA rather than the actual level or accuracy of that SA [99]. Subjective SA ratings have been found not to correlate well with objective measures of SA over several studies, indicating that people have poor awareness of the quality or completeness of their own SA [100]. In order to evaluate SA, a more effective approach is to analyze the performance on the task that has been specifically calibrated for the use case.

6 CONCLUSION

After conducting the evaluation campaign, the conclusions and feedback gathered will serve as a valuable input for refining and iterating the methodology. The objective is to use this methodology to address other use cases and scenarios in the future. Our aim was to pave a way towards a versatile methodology that can be customized to suit various use cases, while being mindful of the fact that we operate in an industrial environment where controlling all variables and using invasive sensors is difficult. Long term and repeated use of AR HMD devices over an extended period are still fully unknown and will remain so because of the lack of ecological validity mentioned above. Some adverse effects, upon which AR HMDs can have a negative impact – are not included in the present document: Perturbation of the circadian cycles', photic seizure etc. . . cannot be measured during global, short, and punctual experimentations, to do so, specific, long, and longitudinal studies will most probably be necessary.

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REFERENCES

- [1] A. D'Anniballe, J. Silva, P. Marzocca, A. Ceruti, The role of augmented reality in air accident investigation and practitioner training, *Reliability Engineering & System Safety*, Volume 204, 2020, 107149, ISSN 0951-8320, <https://doi.org/10.1016/j.ress.2020.107149>. (<https://www.sciencedirect.com/science/article/pii/S0951832020306505>)
- [2] Kristoffer B. Borgen; Timothy D. Ropp; William T. Weldon; (2021). Assessment of Augmented Reality Technology's Impact on Speed of Learning and Task Performance in Aeronautical Engineering Technology Education. *The International Journal of Aerospace Psychology*, (). –. doi :10.1080/24721840.2021.1881403
- [3] M. Hincapi  , A. Caponio, H. Rios, and E. Gonz  lez Mendiv  l, "An introduction to Augmented Reality with applications in aeronautical maintenance," 2011 13th International Conference on Transparent Optical Networks, Stockholm, Sweden, 2011, pp. 1-4, doi: 10.1109/ICTON.2011.5970856. Ceruti *et al.* (2019)
- [4] Ceruti, Alessandro; Marzocca, Pier; Liverani, Alfredo; Bil, Cees (2019). Maintenance in Aeronautics in an Industry 4.0 Context: The Role of Augmented Reality and Additive Manufacturing. *Journal of Computational Design and Engineering*, (). S2288430018302781–. doi: 10.1016/j.jcde.2019.02.001
- [5] Yun Ling *et al.*, "The Relationship between Individual Characteristics and Experienced Presence," *Computers in Human Behavior* 29, no. 4 (2013): 1519–1530.
- [6] James F Knight and Chris Baber, "Effect of Head-Mounted Displays on Posture," *Human factors* 49, no. 5 (2007): 797–807.
- [7] Guill, F. C., & Herd, G. R. (1989). An evaluation of proposed causal mechanisms for" ejection associated" neck injuries. *Aviation, space, and environmental medicine*, 60(7 Pt 2), A26-47.
- [8] McCauley, M. E., & Sharkey, T. J. (1992). Cybersickness: Perception of self-motion in virtual environments. *Presence: Teleoperators & Virtual Environments*, 1(3), 311-318.
- [9] Bouchard, S., Robillard, G., & Renaud, P. (2007). Revising the factor structure of the Simulator Sickness Questionnaire. *Annual review of cybertherapy and telemedicine*, 5(Summer), 128-137.
- [10] Chance, S. S., Gaunet, F., Beall, A. C., & Loomis, J. M. (1998). Locomotion mode affects the updating of objects encountered during travel: The contribution of vestibular and proprioceptive inputs to path integration. *Presence*, 7(2), 168-178.
- [11] Stanney, K., Lawson, B. D., Rokers, B., Dennison, M., Fidopiastis, C., Stoffregen, T., ... & Fulvio, J. M. (2020). Identifying causes of and solutions for cybersickness in immersive technology: reformulation of a research and development agenda. *International Journal of Human-Computer Interaction*, 36(19), 1783-1803.
- [12] Kennedy, R. S., Stanney, K. M., & Dunlap, W. P. (2000). Duration and exposure to virtual environments: sickness curves during and across sessions. *Presence: Teleoperators & Virtual Environments*, 9(5), 463-472
- [13] Natalia Du  ma  ska, Pawe  Strojny, and Agnieszka Strojny, "Can Simulator Sickness Be Avoided? A Review on Temporal Aspects of Simulator Sickness," *Frontiers in Psychology* 9 (2018): 2132, accessed June 4, 2021, <https://doi.org/10.3389/fpsyg.2018.02132>

- [//www.frontiersin.org/article/10.3389/fpsyg.2018.02132/full](https://www.frontiersin.org/article/10.3389/fpsyg.2018.02132/full).
- [14] James Melzer, "Head-Mounted Displays," in *Digital Avionics Handbook*, 3rd ed. (CRC Press, 2015), 22.
- [15] Doyon-Poulin, P. (2014). Effet de l'encombrement visuel de l'écran primaire de vol sur la performance du pilote, la charge de travail et le parcours visuel (Doctoral dissertation, École Polytechnique de Montréal).
- [16] Gish, K. W., & Staplin, L. (1995). Human factors aspects of using head up displays in automobiles: A review of the literature.
- [17] Eriksen, B. A., & Eriksen, C. W. (1974). Effects of noise letters upon the identification of a target letter in a nonsearch task. *Perception & Psychophysics*, 16(1), 143–149. <https://doi.org/10.3758/BF03203267>
- [18] Larish, I. & Wickens, C.D. (1991). Divided attention with superimposed and separated imagery: Implications for head-up displays. Tech Report 91-4, Savoy, IL: ARL)
- [19] Bernd Lorenz, Helmut Tobben, and Sven Schmerwitz, "Human Performance Evaluation of a Pathway HMD," in *Enhanced and Synthetic Vision 2005*, vol. 5802 (Presented at the Enhanced and Synthetic Vision 2005, International Society for Optics and Photonics, 2005), 166–176, accessed July 21, 2021, <https://www.spiedigitallibrary.org/conference-proceedings-of-spie/5802/0000/Human-performance-evaluation-of-a-pathway-HMD/10.1117/12.603280.short>.
- [20] Benjamin J. Dixon *et al.*, "Surgeons Blinded by Enhanced Navigation: The Effect of Augmented Reality on Attention," *Surgical Endoscopy* 27, no. 2 (2013): 454–461, accessed July 20, 2021, <http://link.springer.com/10.1007/s00464-012-2457-3>.
- [21] Anke Huckauf *et al.*, "Perceptual Issues in Optical-See-through Displays," in *Proceedings of the 7th Symposium on Applied Perception in Graphics and Visualization*, 2010, 41–48.
- [22] McCann, R. S., Foyle, D. C., & Johnston, J. C. (1993, June). Attentional limitations with head-up displays. In *Proceedings of the seventh international symposium on aviation psychology* (Vol. 5, No. 01). Ohio State University Columbus.
- [23] Dehais, Frédéric and Fabre, Eve Floriane and Roy, Raphaëlle N. *Cockpit intelligent et interfaces cerveau-machine passives*. (2016) In: *Digital Intelligence 2016*, 4 April 2016 - 6 April 2016 (Laval, Canada). (Unpublished)
- [24] Wolff, M., Vanderhaegen, F., Brethault, M., Brisson, H., & Mollard, R. (2018, October). Vers une possible contribution à la compréhension de l'effet tunnel": une étude exploratoire. In *Proceedings of the Ergo'IA 2018 conference*. Bidart-Biarritz (pp. 3-5).
- [25] Vanderhaegen, F., Wolff, M., & Mollard, R. (2022). A heartbeat-based study of attention in the detection of digital alarms from focused and distributed supervisory control systems. *Cognition, Technology & Work*. doi: 10.1007/s10111-022-00720-4.
- [26] Victor Ng-Thow-Hing *et al.*, "User-Centered Perspectives for Automotive Augmented Reality," in *2013 IEEE International Symposium on Mixed and Augmented Reality - Arts, Media, and Humanities (ISMAR-AMH)* (Presented at the 2013 IEEE International Symposium on Mixed and Augmented Reality - Arts, Media, and Humanities (ISMAR-AMH), Adelaide, Australia: IEEE, 2013), 13–22, accessed July 29, 2021, <http://ieeexplore.ieee.org/document/6671262/>.
- [27] Calvi, A., D'Amico, F., Ferrante, C., & Ciampoli, L. B. (2020). Effectiveness of augmented reality warnings on driving behaviour whilst approaching pedestrian crossings: A driving simulator study. *Accident Analysis & Prevention*, 147, 105760.
- [28] Welford, A. T. (1978). Mental workload as a function of demand, capacity, strategy and skill. *Ergonomics*, 21(3), 151-167.
- [29] Huey, B., & Wickens, C. (1993). Workload factor. *Workload transition: Implications for individual and team performance*, 54-93.
- [30] Hancock, P. A., & Meshkati, N. (Eds.). (1988). *Human mental workload* (pp. 139-183). Amsterdam: North-Holland.
- [31] Gopher, D., & Donchin, E. (1986). *Workload: An examination of the concept*.
- [32] Neville A. Stanton *et al.*, "Seeing through the Mist: An Evaluation of an Iteratively Designed Head-up Display, using a Simulated Degraded Visual Environment, to Facilitate Rotary-Wing Pilot Situation Awareness and Workload," *Cognition, Technology & Work* 22, no. 3 (2020): 549–563, accessed May 27, 2021, <http://link.springer.com/10.1007/s10111-019-00591-2>.
- [33] Franz Viertler and Manfred Hajek, "Evaluation of Visual Augmentation Methods for Rotorcraft Pilots in Degraded Visual Environments," *Journal of the American Helicopter Society* 62, no. 1 (2017): 1–11.
- [34] Bernard Clavier *et al.*, "Vision Augmentée, Vision" Tête Haute" Dans Le Pilotage et Le Contrôle," *I-Revues-Edition Electronique de l'INIST-Information, innovation et interdisciplinarité* (2010): <http://documents>.
- [35] R RD Arents *et al.*, "Predictive Landing Guidance in Synthetic Vision Displays," *The Open Aerospace Engineering Journal* 4, no. 1 (2011).
- [36] Pietro Cipresso *et al.*, "The Past, Present, and Future of Virtual and Augmented Reality Research: A Network and Cluster Analysis of the Literature," *Frontiers in Psychology* 9 (2018), accessed May 12, 2021, <https://www.frontiersin.org/articles/10.3389/fpsyg.2018.02086/full>.
- [37] Luís Fernando de Souza Cardoso, Flávia Cristina Martins Queiroz Mariano, and Ezequiel Roberto Zorral, "A Survey of Industrial Augmented Reality," *Computers & Industrial Engineering* 139 (2020): 106159, accessed May 12, 2021, <https://www.sciencedirect.com/science/article/pii/S036083521930628X>.
- [38] Reitmayr, G., & Schmalstieg, D. (2003, February). Location based applications for mobile augmented reality. In *Proceedings of the Fourth Australasian user interface conference on User interfaces 2003-Volume 18* (pp. 65-73).
- [39] Yeh, M., Wickens, C. D., & Seagull, F. J. (1998). Effects of frame of reference and viewing condition on attentional issues with helmet mounted displays. ILLINOIS UNIV AT URBANA-CHAMPAIGN SAVOYAVIATION RESEARCH LAB.
- [40] Niklas Peinecke *et al.*, "Review of Conformal Displays: More than a Highway in the Sky," *Optical Engineering* 56, no. 5 (2017): 051406, accessed May 25, 2021, [http://opticalengineering.spiedigitallibrary.org/article.aspx?doi=\\$10.1117/1.OE.56.5.051406](http://opticalengineering.spiedigitallibrary.org/article.aspx?doi=$10.1117/1.OE.56.5.051406).
- [41] Harrison, M. F., Forde, K. A., Albert, W. J., Croll, J. C., & Neary, J. P. (2016). Posture and helmet load influences on neck muscle activation. *Aerospace medicine and human performance*, 87(1), 48-53.
- [42] Knight, J. F., & Baber, C. (2004). Neck muscle activity and perceived pain and discomfort due to variations of head load and posture. *Aviation, space, and environmental medicine*, 75(2), 123-131.
- [43] Yan, Y., Chen, K., Xie, Y., Song, Y., & Liu, Y. (2018, July). The effects of weight on comfort of virtual reality devices. In *International Conference on Applied Human Factors and Ergonomics* (pp. 239-248). Springer, Cham.
- [44] Knight, J. F., & Baber, C. (2005). A tool to assess the comfort of wearable computers. *Human factors*, 47(1), 77-91.
- [45] Pearson, E. J. M. (2009). Comfort and its measurement—a literature review. *Disability and Rehabilitation: Assistive Technology*, 4(5), 301-310.
- [46] Ito, K., Tada, M., Ujike, H., & Hyodo, K. (2021). Effects of the Weight and Balance of Head-Mounted Displays on Physical Load. *Applied Sciences*, 11(15), 6802.
- [47] Cattani, G., Andreev, A., Mendoza, C., & Congedo, M. (2018, April). The impact of passive head-mounted virtual reality devices on the quality of EEG signals. In *VRIPHYS 2018-Workshop on Virtual Reality Interaction and Physical Simulation*. The Eurographics Association.
- [48] Hertweck, S., Weber, D., Alwanni, H., Unruh, F., Fischbach, M., Latoschik, M. E., & Ball, T. (2019, March). Brain activity in virtual reality: Assessing signal quality of high-resolution EEG while using head-mounted displays. In *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)* (pp. 970-971). IEEE.
- [49] Kennedy, R. S., Lane, N. E., Berbaum, K. S., and Lilienthal, M. G., "Simulator_Sickness_Questionnaire," *The International Journal of Aviation Psychology*, vol. 3, no. 3, pp. 203–220, 1993.
- [50] Chi, C. F., & Lin, F. T. (1998). A comparison of seven visual fatigue assessment techniques in three data-acquisition VDT tasks. *Human factors*, 40(4), 577-590.
- [51] Kim Y. Y., Kim H. J., Kim E. N., Ko H. D., Kim H. T. (2005). Characteristic changes in the physiological components of cybersickness. *Psychophysiology* 42(5):616–625
- [52] Nam, Y., Hong, U., Chung, H., & Noh, S. R. (2021). Eye Movement Patterns Reflecting Cybersickness: Evidence from Different Experience Modes of a Virtual Reality Game. *Cyberpsychology, Behavior, and Social Networking*.
- [53] Wibirama S., Santosa P. I., Widyarani P., Brilianto N., Hafidh W. (2020). Physical discomfort and eye movements during arbitrary and optical flow-like motions in stereo 3d contents. *Virtual Reality*, 24(1), 39–51.
- [54] Fukuda, K., Stern, J. A., Brown, T. B., & Russo, M. B. (2005). Cognition, blinks, eye-movements, and pupillary movements during performance of a running memory task. *Aviation, space, and environmental medicine*, 76(7), C75-C85.
- [55] Stern, J. A., Boyer, D., & Schroeder, D. (1994). Blink rate: a possible measure of fatigue. *Human factors*, 36(2), 285-297.
- [56] Blehm, C., Vishnu, S., Khattak, A., Mitra, S., & Yee, R. W. (2005). Computer vision syndrome: a review. *Survey of ophthalmology*, 50(3), 253-262.T
- [57] Rosenfield, M. (2011). Computer vision syndrome: a review of ocular causes and potential treatments. *Ophthalmic and Physiological Optics*, 31(5), 502-515.
- [58] Sheedy, J. E., Gowrisankaran, S., & Hayes, J. R. (2005). Blink rate decreases with eyelid squint. *Optometry and vision science*, 82(10), 905-911.
- [59] Sheppard, A. L., & Wolffsohn, J. S. (2018). Digital eye strain: prevalence, measurement and amelioration. *BMJ open ophthalmology*, 3(1), e000146.
- [60] Kumashiro, M. (1995). Practical measurement of psycho-physiological functions for determining workloads. *Evaluation of Human Work: A Practical Ergonomics Methodology*. Philadelphia: Taylor & Francis.
- [61] Eisen-Enosh, A., Farah, N., Burgansky-Eliash, Z., Polat, U., & Mandel, Y. (2017). Evaluation of critical flicker-fusion frequency measurement methods for the investigation of visual temporal resolution. *Scientific reports*,
- [62] Rosenholtz, R., Li, Y., & Nakano, L. (2007). Measuring visual clutter. *Journal of vision*, 7(2), 17-17
- [63] Viertler, F., Krammer, C., & Hajek, M. (2015). Analyzing visual clutter of 3D-conformal HMD solutions for rotorcraft pilots in degraded visual environment.
- [64] Kaber, D., Kim, S.-H., Kaufmann, K., Alexander, A., Steltzer, E., Hsiang, S. (2008). Modeling the effects of HUD visual properties, pilot experience and flight scenario on a multi-dimensional measure of clutter. Hampton, VA: Langley Research Center.

- [65] Kim, S.-H., Prinzel, L. J., Kaber, D. B., Alexander, A. L., Stelzer, E. M., Kaufmann, K., Veil, T. (2011). Multidimensional measure of display clutter and pilot performance for advanced head-up display. *Aviation, Space, and Environmental Medicine*, 82, 1013–1022.
- [66] Moacdieh, N., & Sarter, N. (2015). Display clutter: A review of definitions and measurement techniques. *Human factors*, 57(1), 61–100.
- [67] Posner, M.I. 1980. Orienting of attention. *Quarterly Journal of Experimental Psychology*, 32, 3–25.
- [68] Corbetta M, Akbudak E, Conturo TE, Snyder AZ, Ollinger JM, Drury HA, et al. A common network of functional areas for attention and eye movements. *Neuron*. 1998 Oct;21(4):761–73. PMID:9808463
- [69] Deubel H, Schneider WX. Saccade target selection and object recognition: evidence for a common attentional mechanism. *Vision Res*. 1996 Jun;36(12):1827–37. PMID:8759451
- [70] Thomas, L.C., & Wickens C.D. (2004). Eye tracking and individual differences in off-normal event detection when flying with a synthetic vision system display. In *Human Factors and Ergonomics Society Annual Meeting Proceedings*, 48 (pp. 223–227). HFES, Santa Monica, California, USA.
- [71] Goldberg, J.H., & Kotval X.P. (1999). Computer interface evaluation using eye movements: Methods and constructs. *International Journal of Industrial Ergonomics* 24, 631–645.
- [72] Rayner, K. (1998). Eye movements in reading and information processing: 20 years of research. *Psychological bulletin*, 124(3), 372–422.
- [73] Dehais, F., Behrend, J., Peysakhovich, V., Causse, M., & Wickens, C. D. (2017). Pilot Flying and Pilot Monitoring's Aircraft State Awareness During Go-Around Execution in Aviation: A Behavioral and Eye Tracking Study. *The International Journal of Aerospace Psychology*, 27(1-2), 15–28.
- [74] Mackenzie, A. K., & Harris, J. M. (2015). Eye movements and hazard perception in active and passive driving. *Visual cognition*, 23(6), 736–757.
- [75] Recarte, M. A., & Nunes, L. M. (2003). Mental workload while driving: effects on visual search, discrimination, and decision making. *Journal of experimental psychology: Applied*, 9(2), 119.
- [76] Wang Y, Reimer B, Dobres J, Mehler B. The sensitivity of different methodologies for characterizing drivers' gaze concentration under increased cognitive demand. *Transp Res Part F Traffic Psychol Behav*. 2014 Sep 1;26:227–37.
- [77] Wolff, M., Vanderhaegen, F., Brethault, M., Brisson, H., & Mollard, R. (2018, October). Towards a contribution to understanding" the tunnel effect" an exploratory study. In *Proceedings of the 16th Ergo'IA "Ergonomie Et Informatique Avancée" Conference* (pp. 1-10).
- [78] Vanderhaegen, F., Wolff, M., & Mollard, R. (2019). Heart-Computer Synchronization Interface to control human-machine symbiosis: a new human availability support for cooperative systems. *IFAC-PapersOnLine* 52(19),91-96. doi: 10.1016/j.ifacol.2019.12.116
- [79] Selcon, S. J., & Taylor, R. M. (1990). Evaluation of the Situational Awareness Rating Technique (SART) as a tool for aircrew systems design. *AGARD, Situational Awareness in Aerospace Operations* 8 p(SEE N 90-28972 23-53).
- [80] Hart, S. G., & Staveland, L. E. (1988). Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research. In *Advances in psychology* (Vol. 52, pp. 139-183). North-Holland.
- [81] Hart, S. G. (2006, October). NASA-task load index (NASA-TLX); 20 years later. In *Proceedings of the human factors and ergonomics society annual meeting* (Vol. 50, No. 9, pp. 904-908). Sage CA: Los Angeles, CA: Sage publications.
- [82] Veiersted, K. B. (1991). The reproducibility of test contractions for calibration of electromyographic measurements. *European journal of applied physiology and occupational physiology*, 62, 91-98.
- [83] Chihara, T., & Seo, A. (2018). Evaluation of physical workload affected by mass and center of mass of head-mounted display. *Applied ergonomics*, 68, 204-212.
- [84] González-Izal, M., Malanda, A., Gorostiaga, E., & Izquierdo, M. (2012). Electromyographic models to assess muscle fatigue. *Journal of Electromyography and Kinesiology*, 22(4), 501-512.
- [85] Hermens, H., & Freriks, B. (1997). The state of the art on sensors and sensor placement procedures for surface electromyography: a proposal for sensor placement procedures. Enschede, The Netherlands: Roessingh Research and Development. Basmajian & De Luca (1985)
- [86] Ayachi, F. S. (2011). Etude du recrutement des unités motrices par analyse du signal EMG de surface (Doctoral dissertation, Compiègne).
- [87] Ehinger, B. V., Groß, K., Ibs, I., & König, P. (2019). A new comprehensive eye-tracking test battery concurrently evaluating the Pupil Labs glasses and the EyeLink 1000. *PeerJ*, 7, e7086.
- [88] Drouot, M., Le Bigot, N., Bricard, E., De Bougrenet, J. L., & Nourrit, V. (2022). Augmented reality on industrial assembly line: Impact on effectiveness and mental workload. *Applied Ergonomics*, 103, 103793.
- [89] Wolff, M., Vanderhaegen, F., Brethault, M., Brisson, H., & Mollard, R. (2018, October). Towards a contribution to understanding" the tunnel effect" an exploratory study. In *Proceedings of the 16th Ergo'IA "Ergonomie Et Informatique Avancée" Conference* (pp. 1-10).
- [90] Shen, W., & Parsons, K. C. (1997). Validity and reliability of rating scales for seated pressure discomfort. *International journal of industrial ergonomics*, 20(6), 441-461.
- [91] Benedetto, S., Draï-Zerbib, V., Pedrotti, M., Tissier, G., & Baccino, T. (2013). E-readers and visual fatigue. *PLoS one*, 8(12), e83676.
- [92] Rossignol, A. M., Morse, E. P., Summers, V. M., & Pagnotto, L. D. (1987). Video display terminal use and reported health symptoms among Massachusetts clerical workers. *Journal of Occupational medicine*, 112-118.
- [93] Blehm, C., Vishnu, S., Khattak, A., Mitra, S., & Yee, R. W. (2005). Computer vision syndrome: a review. *Survey of ophthalmology*, 50(3), 253-262.T
- [94] Xie, X., Song, F., Liu, Y., Wang, S., & Yu, D. (2021). Study on the effects of display color mode and luminance contrast on visual fatigue. *IEEE Access*, 9, 35915-35923.
- [95] Cardona, G., García, C., Serés, C., Vilaseca, M., & Gispets, J. (2011). Blink rate, blink amplitude, and tear film integrity during dynamic visual display terminal tasks. *Current eye research*, 36(3), 190-197.
- [96] Faulhaber, A. K., & Friedrich, M. (2019, November). Eye tracking metrics as an indicator of workload in commercial single-pilot operations. In *International Symposium on Human Mental Workload: Models and Applications* (pp. 213-225). Springer, Cham.
- [97] Kramer, L. J., Etherington, T. J., Last, M. C., Kennedy, K. D., & Bailey, R. E. (2017, September). Quantifying pilot contribution to flight safety during drive shaft failure. In *2017 IEEE/AIAA 36th Digital Avionics Systems Conference (DASC)* (pp. 1-7). IEEE.
- [98] Guastello, S. J., Marra, D. E., Corroero, A. N., Michels, M., & Schimmel, H. (2017, June). Elasticity and rigidity constructs and ratings of subjective workload for individuals and groups. In *International Symposium on Human Mental Workload: Models and Applications* (pp. 51-76). Springer, Cham.
- [99] Endsley, M. R., Selcon, S. J., Hardiman, T. D., & Croft, D. G. (1998, October). A comparative analysis of SAGAT and SART for evaluations of situation awareness. In *Proceedings of the human factors and ergonomics society annual meeting* (Vol. 42, No. 1, pp. 82-86). Sage CA: Los Angeles, CA: SAGE Publications.
- [100] Endsley, M. R. (2020a). The divergence of objective and subjective situation awareness: A meta-analysis. *Journal of Cognitive Engineering and Decision Making*, 14(1), 34–53.